



Remote health monitoring system to combat Covid-19

All India Institute of Medical Sciences (AIIMS), Rishikesh, in collaboration with Navratna Defence PSU Bharat Electronics Limited (BEL), has developed a state-of-the-art health monitoring system to remotely assess the health of Covid-19 suspects/patients quarantined in homes and hospitals. The solution aims to significantly reduce the risk of exposure to healthcare workers. It is also expected to reduce the increasing demand of PPE and other logistics.

Due to the outbreak of Covid-19 pandemic, the need to develop a solution to remotely monitor patients for limiting exposure of healthcare workers was prime and urgent. To combat this need, AIIMS-Rishikesh and BEL came together to provide a comprehensive digital and clinically appropriate solution. Based on inputs given by AIIMS-Rishikesh, BEL

developed the Proof of Concept (PoC) model of a system integrating non-invasive health monitoring sensors to measure critical parameters such as temperature, pulse rate, SPO₂ (saturated oxygen level) and respiration rate.

A mobile app/Web browser has been developed for people to get enrolled with AIIMS-Rishikesh once they show symptoms of Covid-19. AIIMS-Rishikesh will study patient complaints and based on the assessment by clinical experts, a health monitoring kit will be handed over to the patient for periodical monitoring of the critical parameters. Software will give out alerts in the form of messages to medical officers and healthcare workers when the health parameters exceed the threshold. It will also record the severity of the patient's condition in different colour codes.

VersaTouch brings touch localisation and force sensing to everyday surfaces

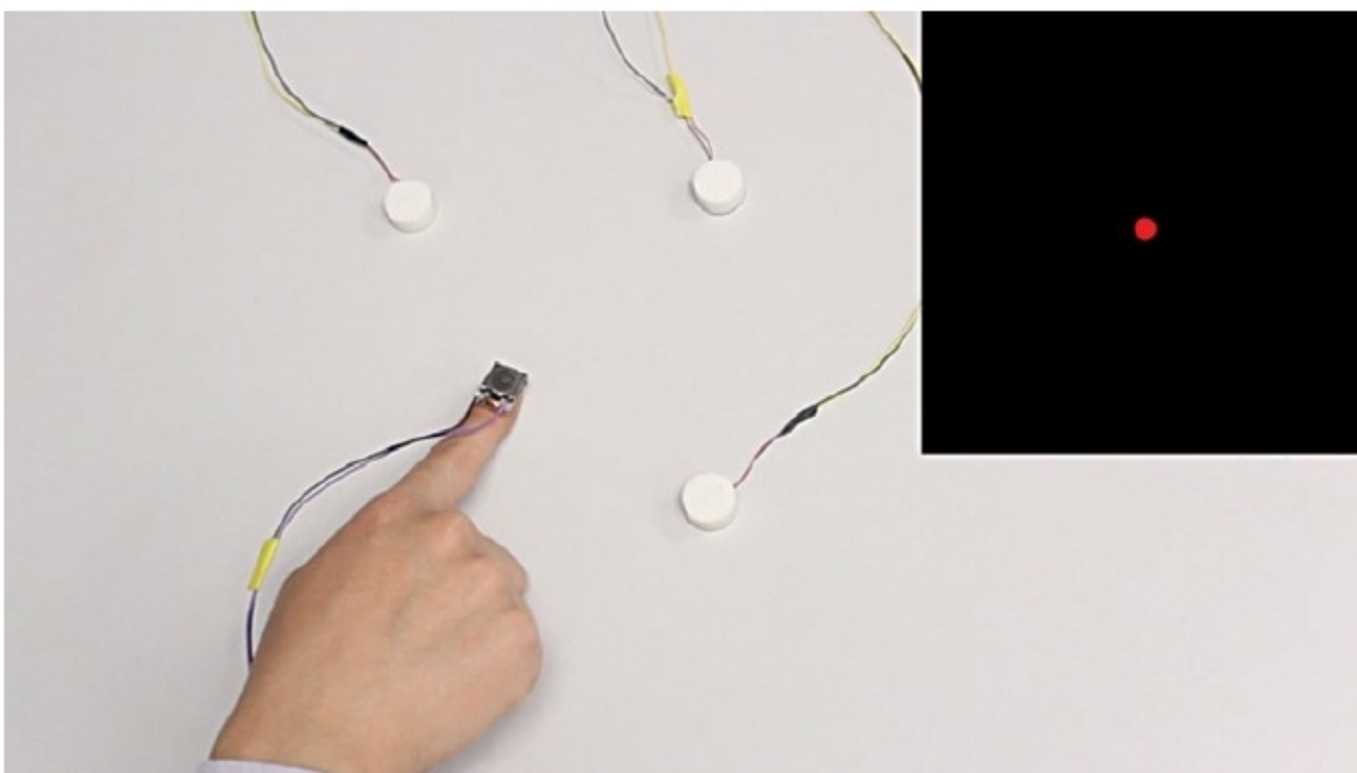
Researchers from the University of Auckland in New Zealand's Augmented Human Lab are exploring a new way to construct interactive touch surfaces, using finger-mounted audio transducers.

VersaTouch—which works on everyday surfaces—uses

one or more receivers to measure sound waves emanating from the wearer's 'augmented' fingers, allowing it to calculate their positions and/or movements. The plug-and-play system can also sense force based on a changing audio signature and track individual digits by alternating each one's sonic outputs.

Importantly, VersaTouch can be configured without permanent modification to the newly interactive surface. The setup includes an Arduino Due to receive signals, a Teensy 3.6 to control the transducers, and a MacBook to process the data and calculate the touch positions with a Java program.

It is a portable, plug-and-play system that uses active acoustic sensing to track fine-grained touch locations as well as touch force of multiple fingers on everyday surfaces without having to permanently instrument them or do an extensive calibration.



Representational image